

Central limit theorem and confidence intervals

Class 6

Objective:

Understand the Central limit theorem and one of its most useful implications: confidence intervals

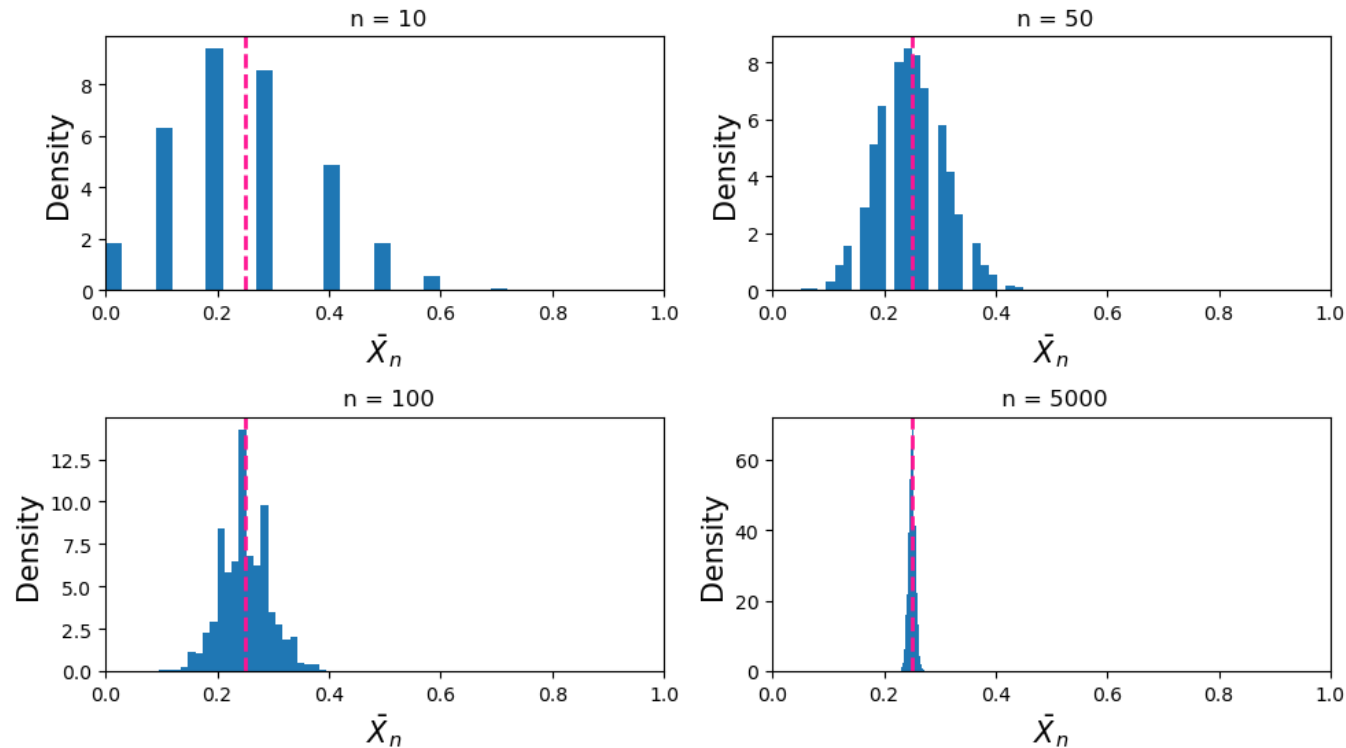
Central limit theorem

Examining the convergence of the mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

Bernoulli(p=1/4): distribution of the sample mean

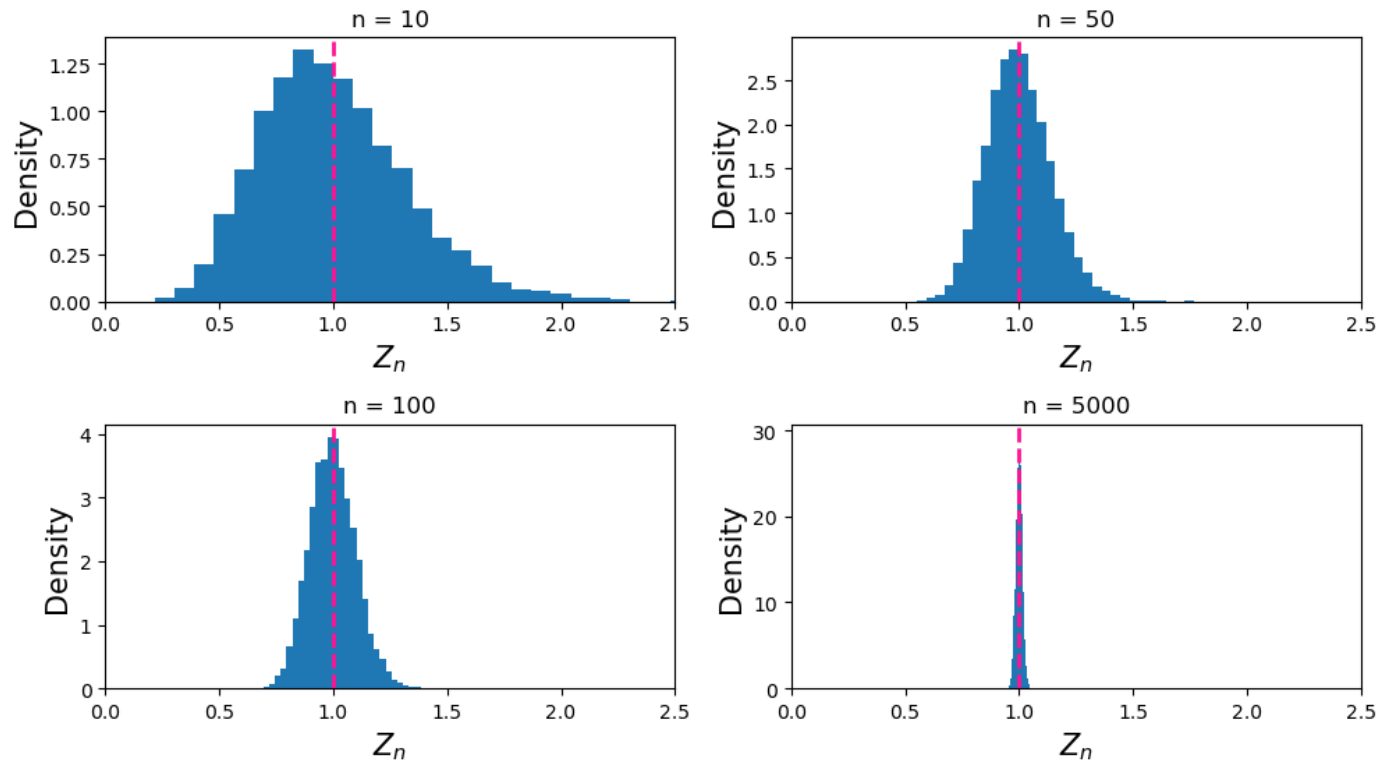


Examining the convergence of the mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$X \sim \text{Exponential}(1)$$

Exponential(lambda=1): distribution of the sample mean



Normalizing the mean estimator

Let $X_i, i \in \{1, \dots, n\}$ be identically distributed random variables and $\mathbb{E}[X_1] = \mu$
 $\text{Var}(X_1) = \sigma^2$. Then:

$$\mathbb{E}[\bar{X}_n] = \mu \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

Consider the random variable:

$$Z_n = \frac{\bar{X}_n - \mu}{\sqrt{\frac{\sigma^2}{n}}} \quad \text{Then: } \mathbb{E}[Z_n] = 0 \text{ and } \text{Var}(Z_n) = 1$$

Z_n can be written as:

$$Z_n = \frac{\frac{X_1 + X_2 + \dots + X_n}{n} - \mu}{\sqrt{\frac{\sigma^2}{n}}} = \frac{\frac{X_1 + X_2 + \dots + X_n - n\mu}{n}}{\frac{\sigma}{\sqrt{n}}} = \frac{X_1 + X_2 + \dots + X_n - n\mu}{\sigma\sqrt{n}}$$

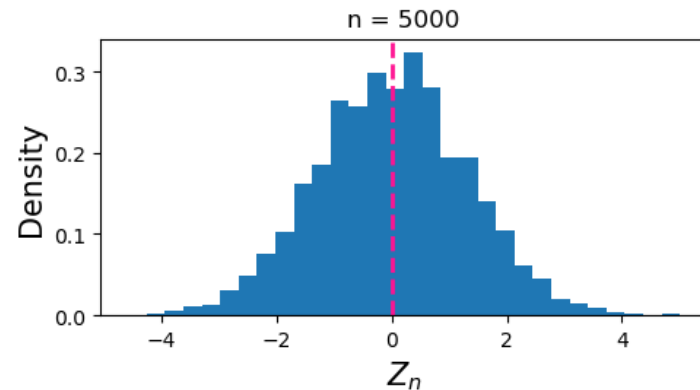
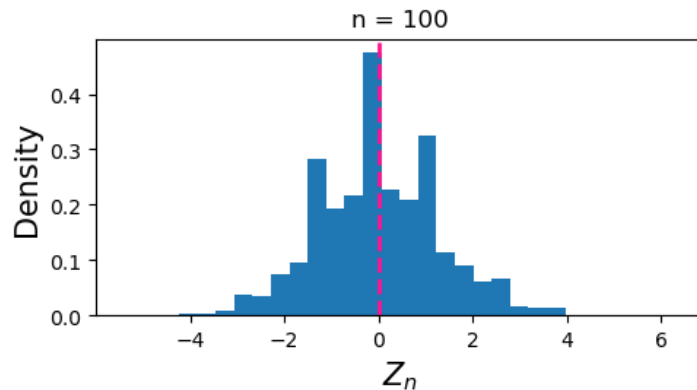
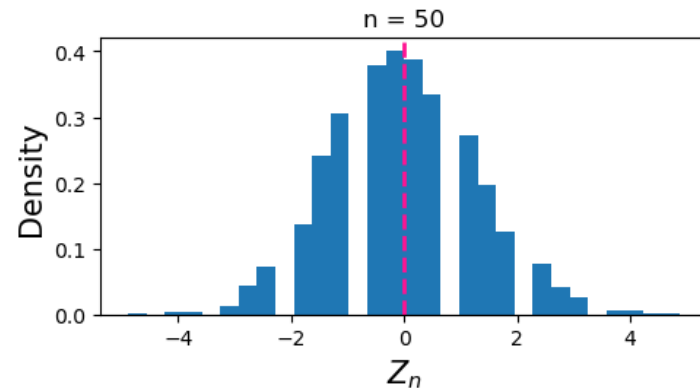
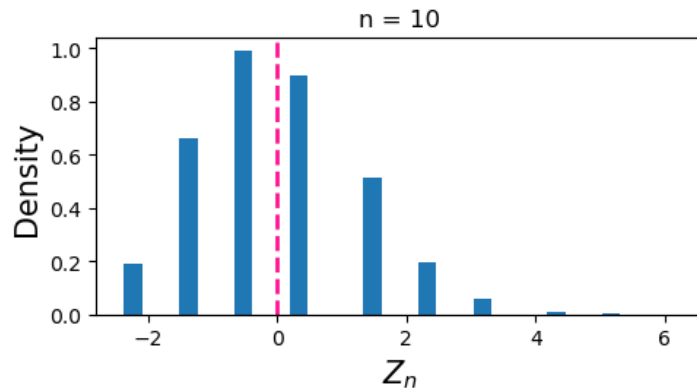
$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

Visual normalization of the mean

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

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Bernoulli(p=1/4): distribution of the sample mean

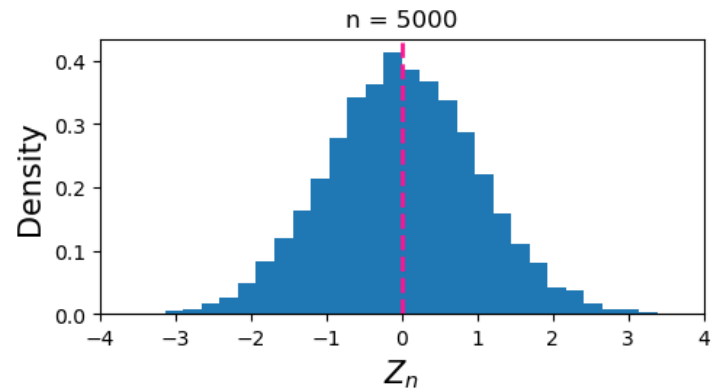
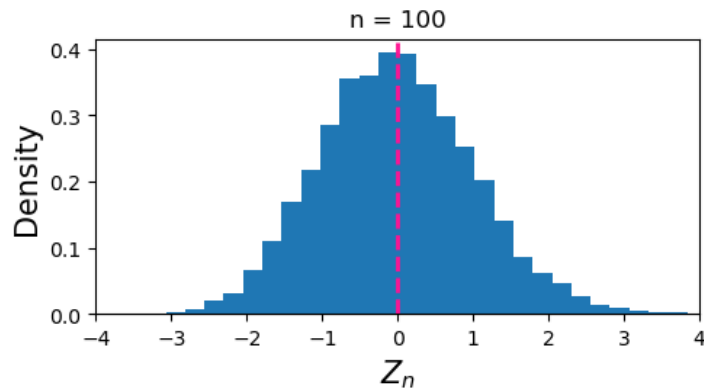
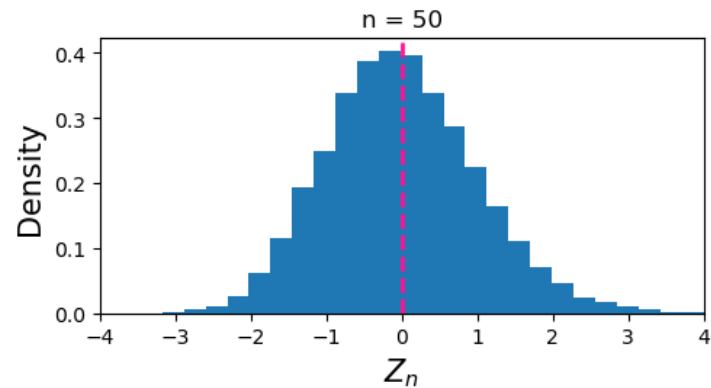
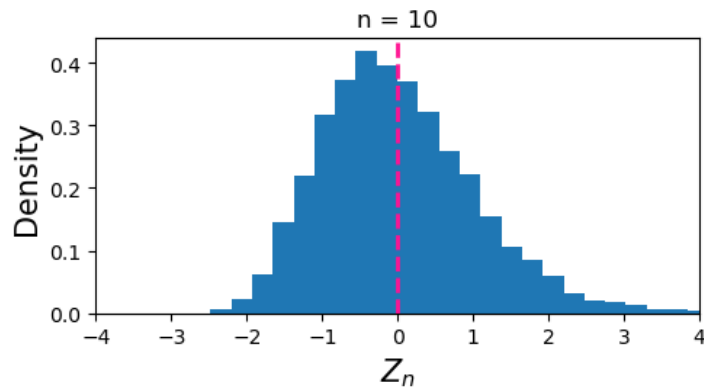


Visual normalization of the mean

$$X \sim \text{Exponential}(1)$$

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

Exponential(lambda = 1): distribution of the sample mean



Central limit theorem

The theorem:

Let $X_1, X_2, \dots$ be independent and identically distributed random variables with mean $\mu := \mathbb{E}[X_1]$ and variance $\sigma^2 = \text{Var}[X_1] < \infty$. Define the standardized sum:

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}}$$

Then, as $n \rightarrow \infty$, the distribution of Z_n converges to the standard normal distribution:

$$Z_n \xrightarrow{d} \mathcal{N}(0, 1)$$

Discussion:

- What should it mean the distribution convergence $\xrightarrow{d}$?

Convergence in distribution

Definition:

The sequence of random variables Z_n converges in distribution to $Z \sim \mathcal{N}(0, 1)$, if:

$$\lim_{n \rightarrow \infty} F_{Z_n}(t) = F_Z(t) \quad \forall t \in \mathbb{R}$$

In this case for all set $A \subset \mathbb{R}$ « sufficiently regular »:

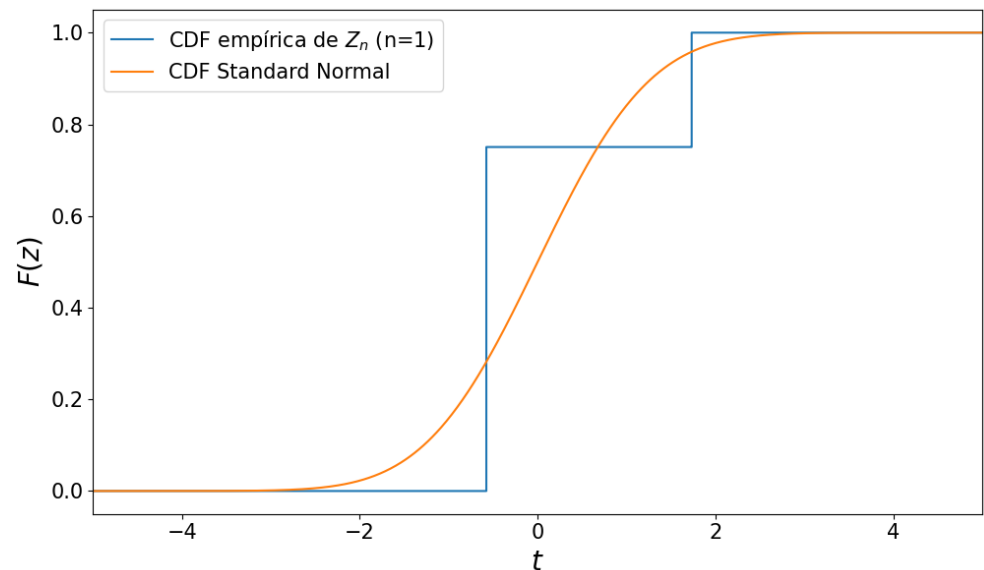
$$P(Z_n \in A) \rightarrow P(Z \in A) \rightarrow A \text{ can be any interval}$$

We denote: $Z_n \xrightarrow{d} Z$

Exemple:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 1$



Convergence in distribution

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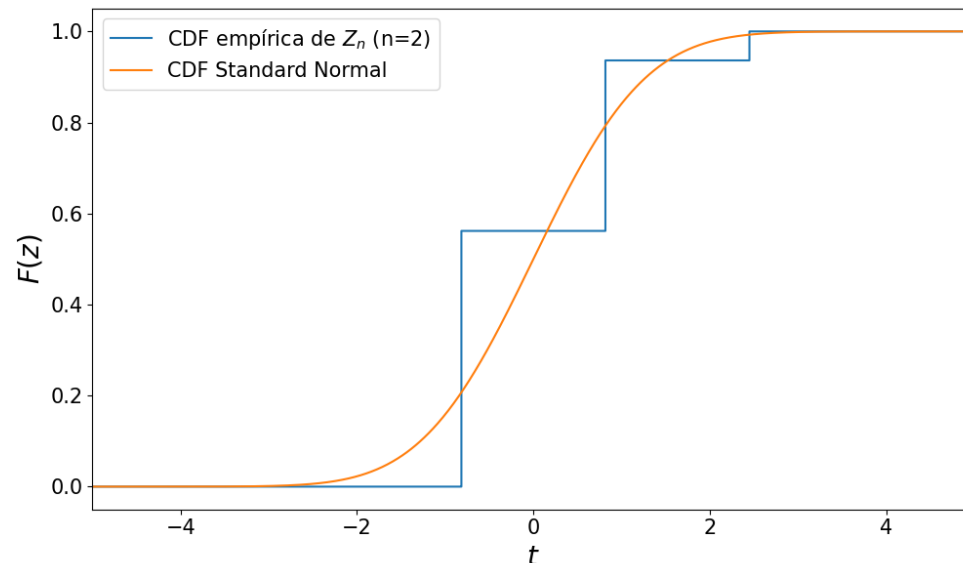
$$P(Z_n \in A) \longrightarrow P(Z \in A)$$

We denote: $Z_n \xrightarrow{d} Z$

Exemple:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 2$



Convergence in distribution

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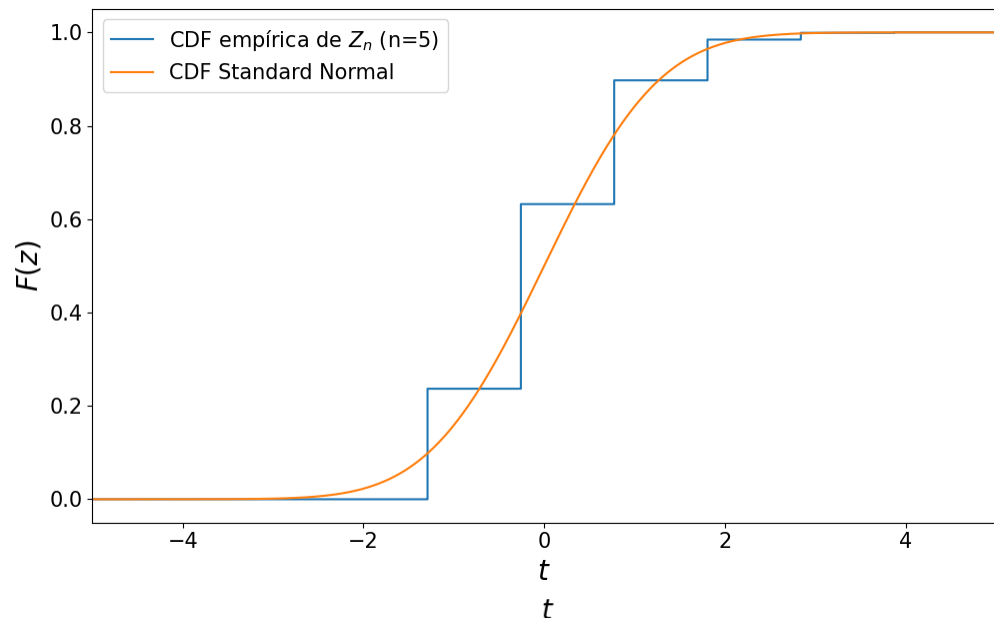
$$P(Z_n \in A) \longrightarrow P(Z \in A)$$

We denote: $Z_n \xrightarrow{d} Z$

Example:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 5$



Convergence in distribution

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The sequence of random variables Z_n converges in distribution to $Z \sim \mathcal{N}(0, 1)$, if:

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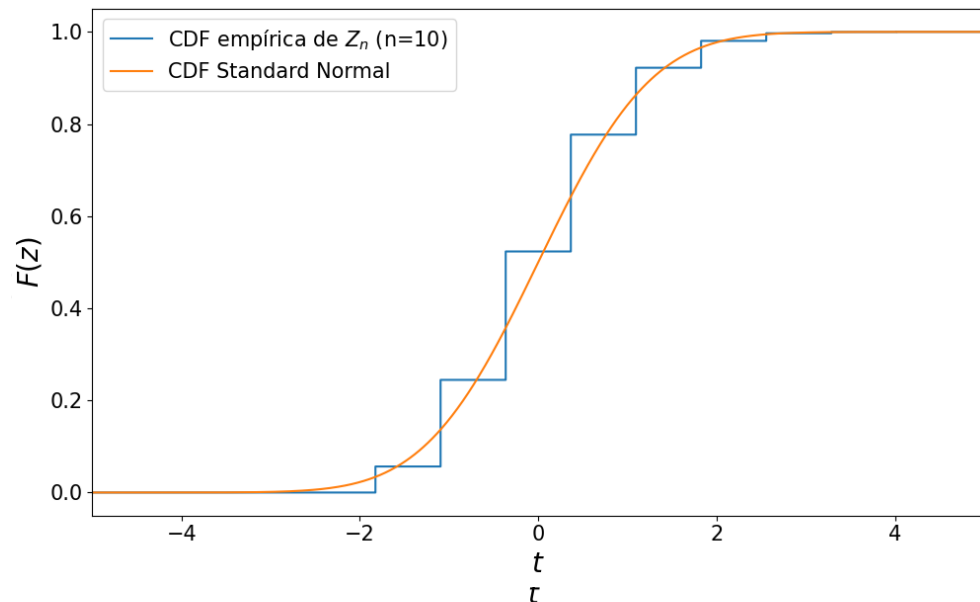
$$P(Z_n \in A) \longrightarrow P(Z \in A)$$

We denote: $Z_n \xrightarrow{d} Z$

Exemple:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 10$



Convergence in distribution

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The sequence of random variables Z_n converges in distribution to $Z \sim \mathcal{N}(0, 1)$, if:

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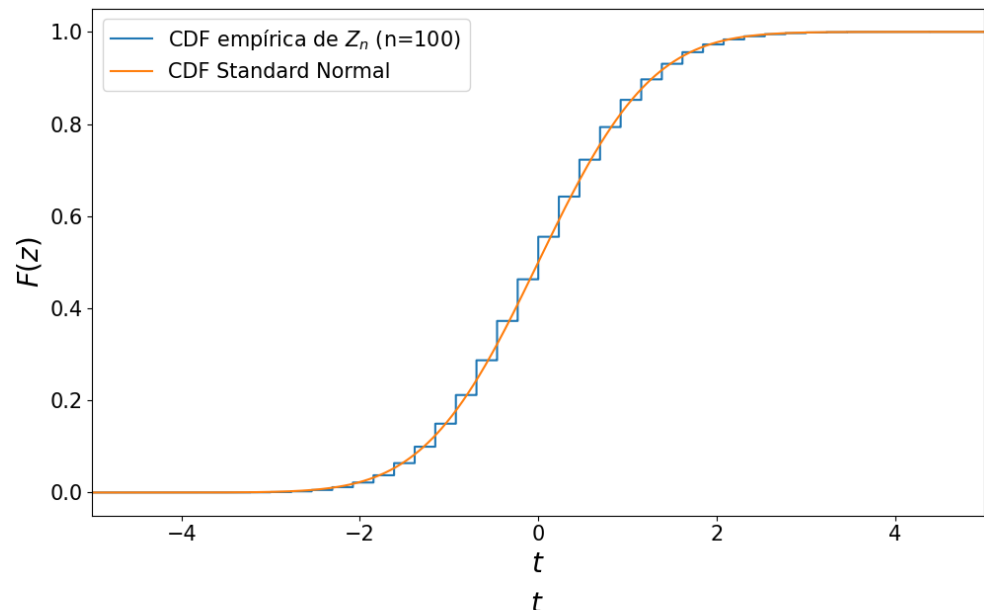
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We denote: $Z_n \xrightarrow{d} Z$

Example:

$X_i \sim \text{Bernoulli}(0.25)$, i.i.d

$n = 100$



Confidence intervals

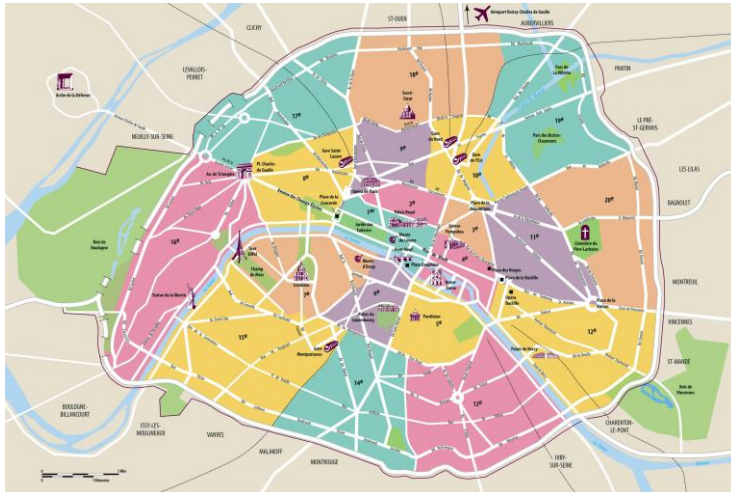
Practical application: All empirical science!

All empirical sciences (physics, biology, climatology) aim to infer something about an unknown reality from data. The natural next question is: given the data I have, what are the plausible values of the true underlying quantity?

Election pool

A change in perspective: So far, we have visualized and analyzed some useful measures to understand data. But what if these data represent not an entire population, but only a sample?

Example: Election pool



Candidate	Number of respondents
Candidate A	362
Candidate B	638

Table 2: Election poll results with 1,000 respondents.

Statistical question: How to estimate the real result?

Model for election pool

Probabilistic model

Let p be the **true** (unknown) proportion of people who vote for candidate A, and $1-p$ the proportion who vote for candidate B. Based on the survey, we aim to **estimate** p .

Therefore, we have a Bernoulli random variable with parameter p :

$$X = \begin{cases} 1, & \text{if the person votes for candidate A,} \\ 0, & \text{if the person votes for candidate B.} \end{cases}$$

Data

The survey is viewed as a **realization** or **outcome** $(x_1, x_2, \dots, x_{1000})$ of the random vector $(X_1, X_2, \dots, X_{1000})$, where $X_i \sim \text{Bernoulli}(p), i \in \{1, \dots, 1000\}$

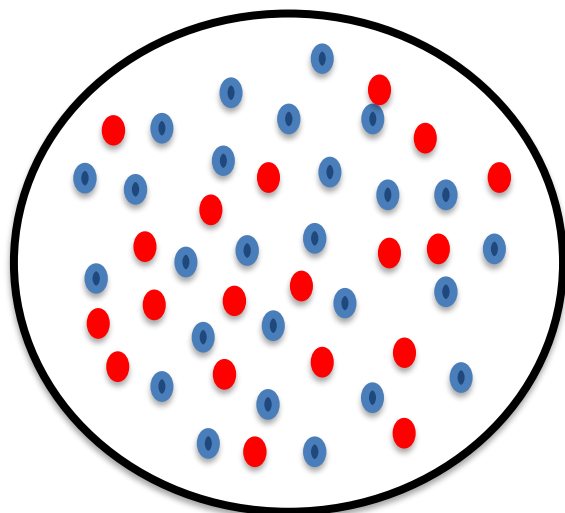
What are we assuming with this model?

- All respondents are independent
- The value of p is fixed during the survey
- Each respondent is chosen at random



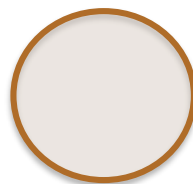
Model for election pool

Population



● Candidate A
● Candidate B

Survey



Model

$$X = \begin{cases} 1, & \text{Candidate A,} \\ 0, & \text{Candidate B.} \end{cases}$$

$$X_i \sim \text{Bernoulli}(p), \quad i \in \{1, \dots, n\}$$

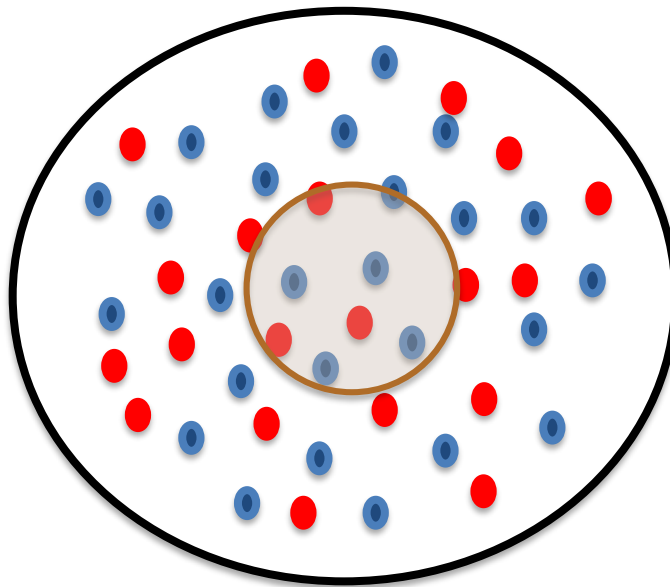
Data: $(x_1, \dots, x_n)$

Data mean: $\bar{x}_n$

Possible outcomes of the mean estimator

Population

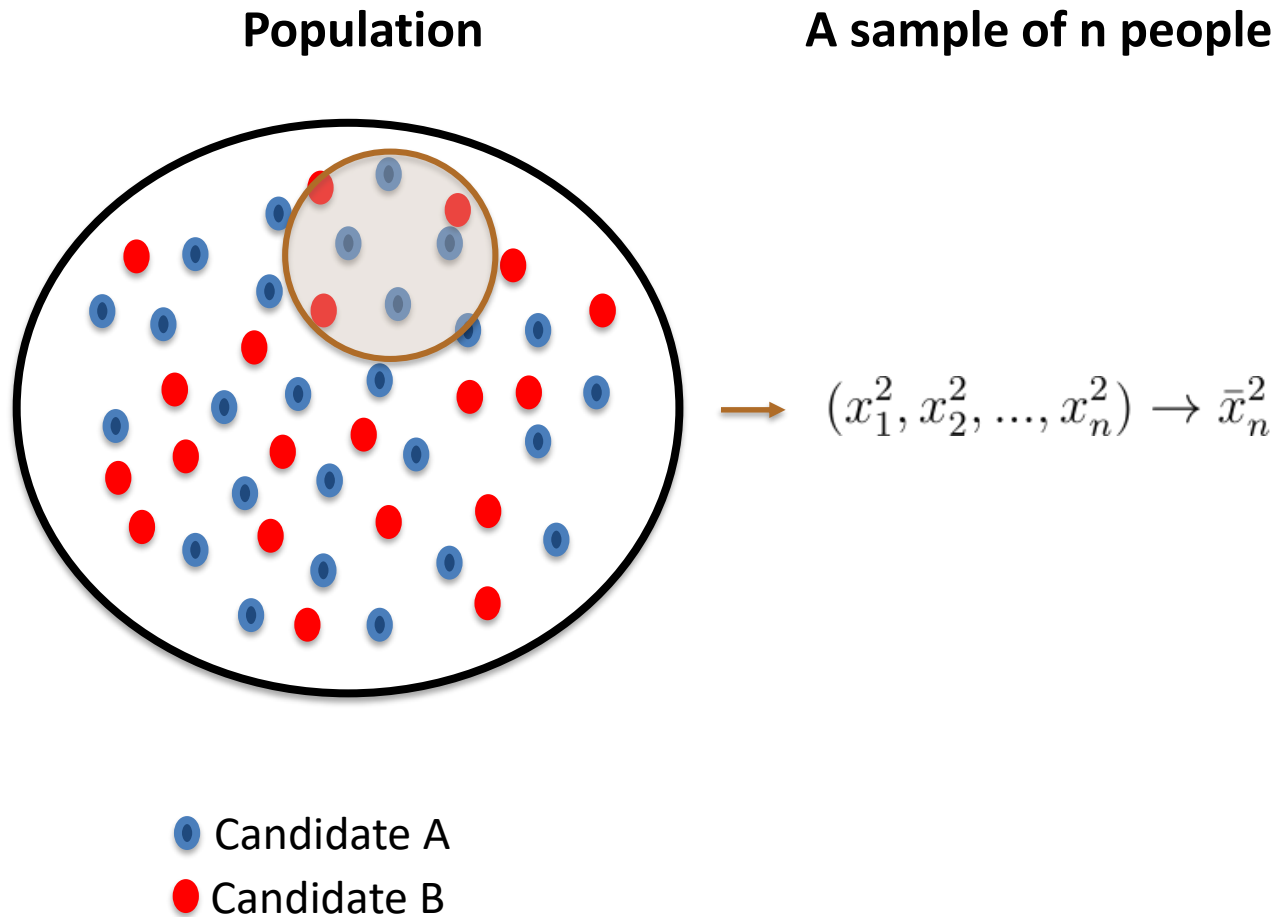
A sample of n people



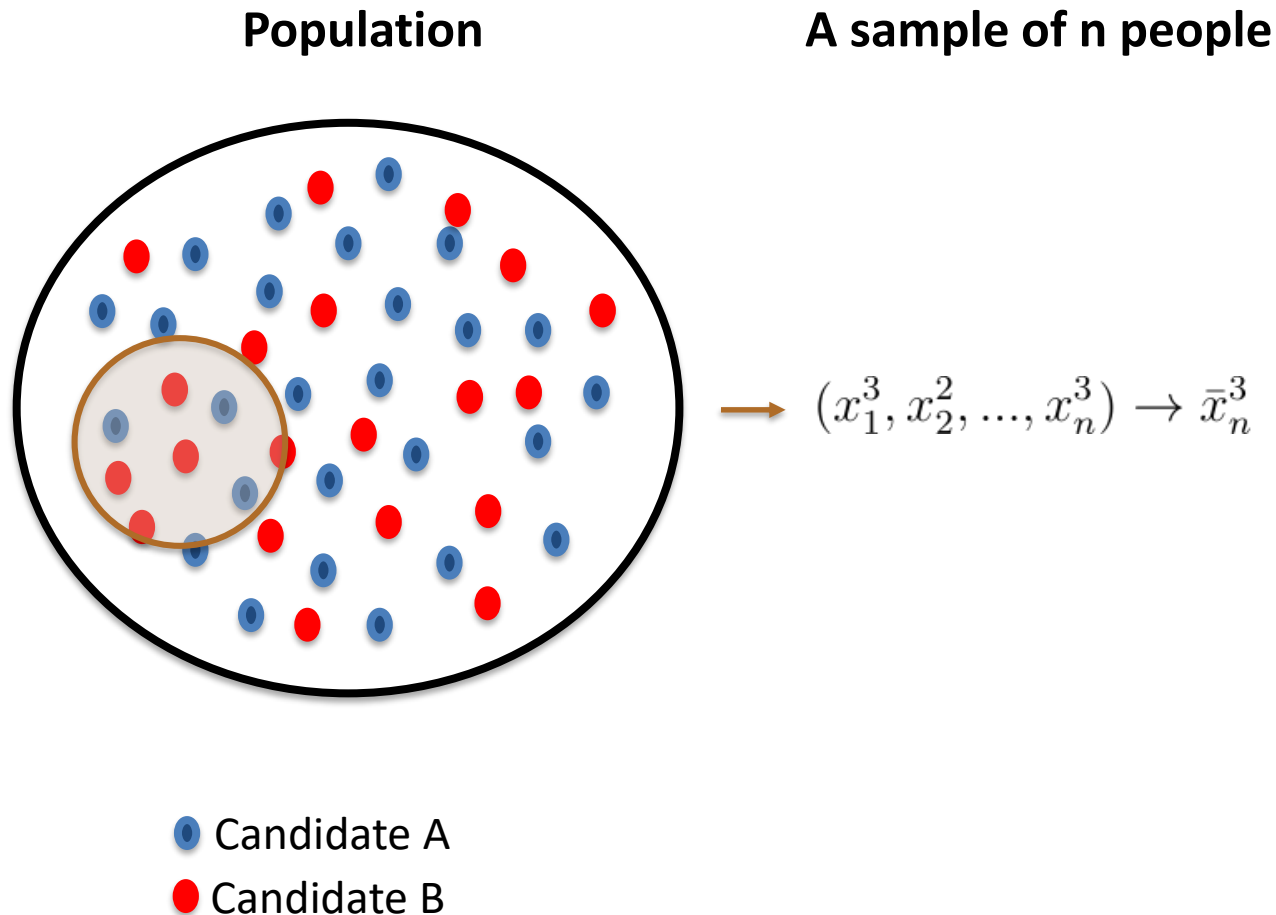
$$\rightarrow (x_1^1, x_2^1, \dots, x_n^1) \rightarrow \bar{x}_n^1$$

● Candidate A
● Candidate B

Possible outcomes of the mean estimator



Possible outcomes of the mean estimator



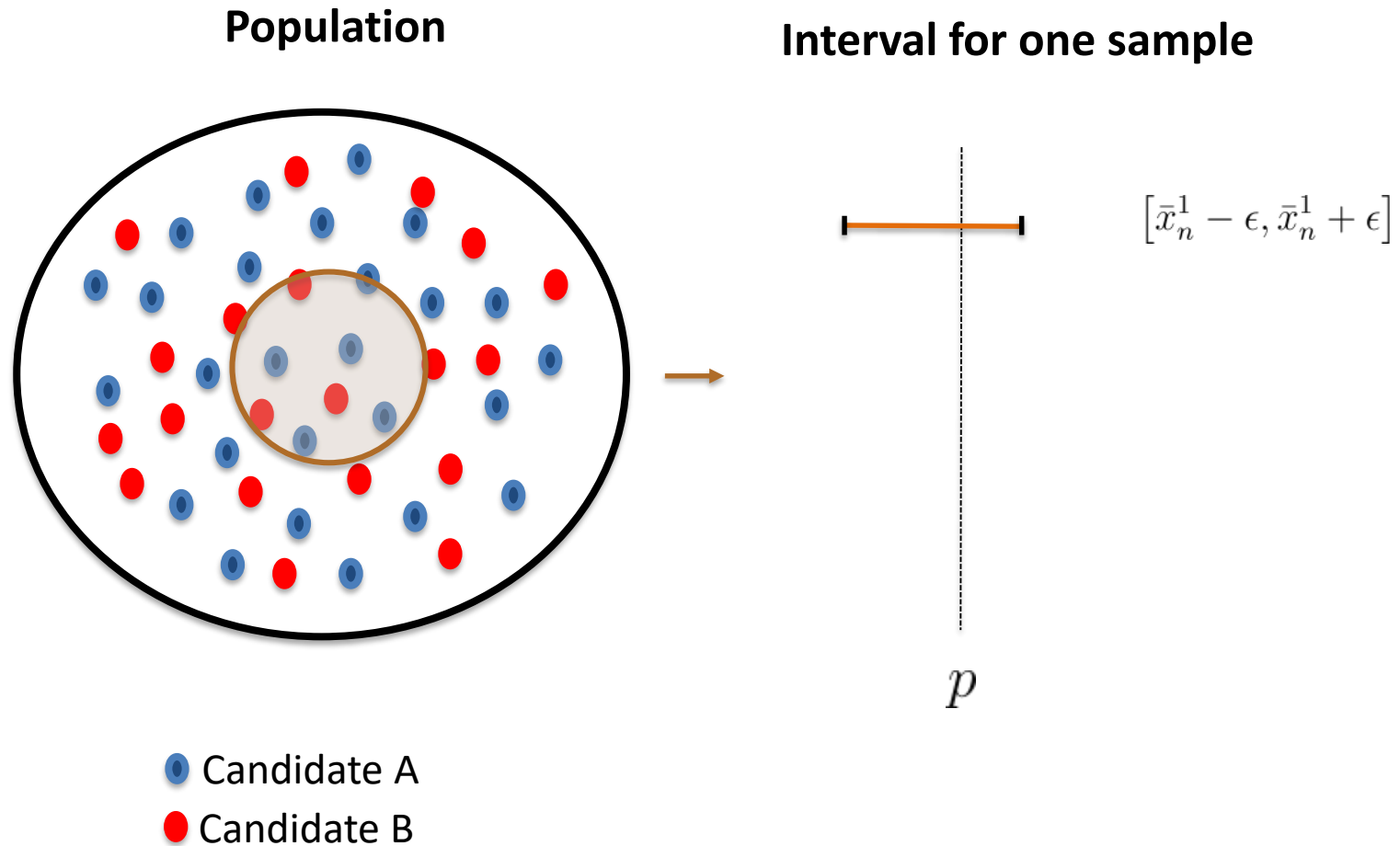
A random interval containing p

- $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ is a random variable
- Our survey is one realization (or outcome) of $\bar{X}_n$
- We can not expect to estimate p as a fixed number
- We would like to consider a margin of error for our survey

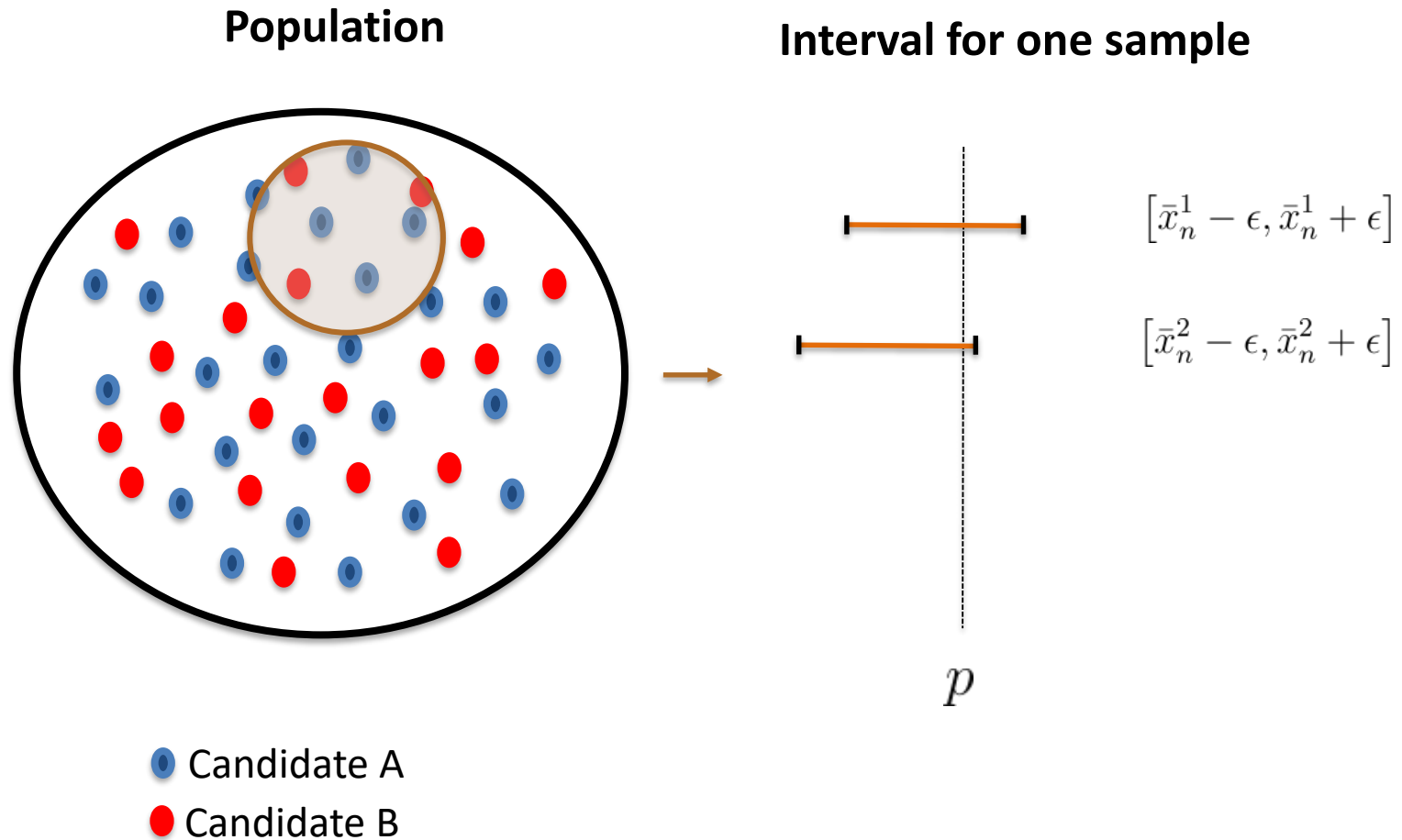
Suppose that we want to find a margin of error $\epsilon > 0$ such that:

$$p \in [\bar{x}_n - \epsilon, \bar{x}_n + \epsilon]$$

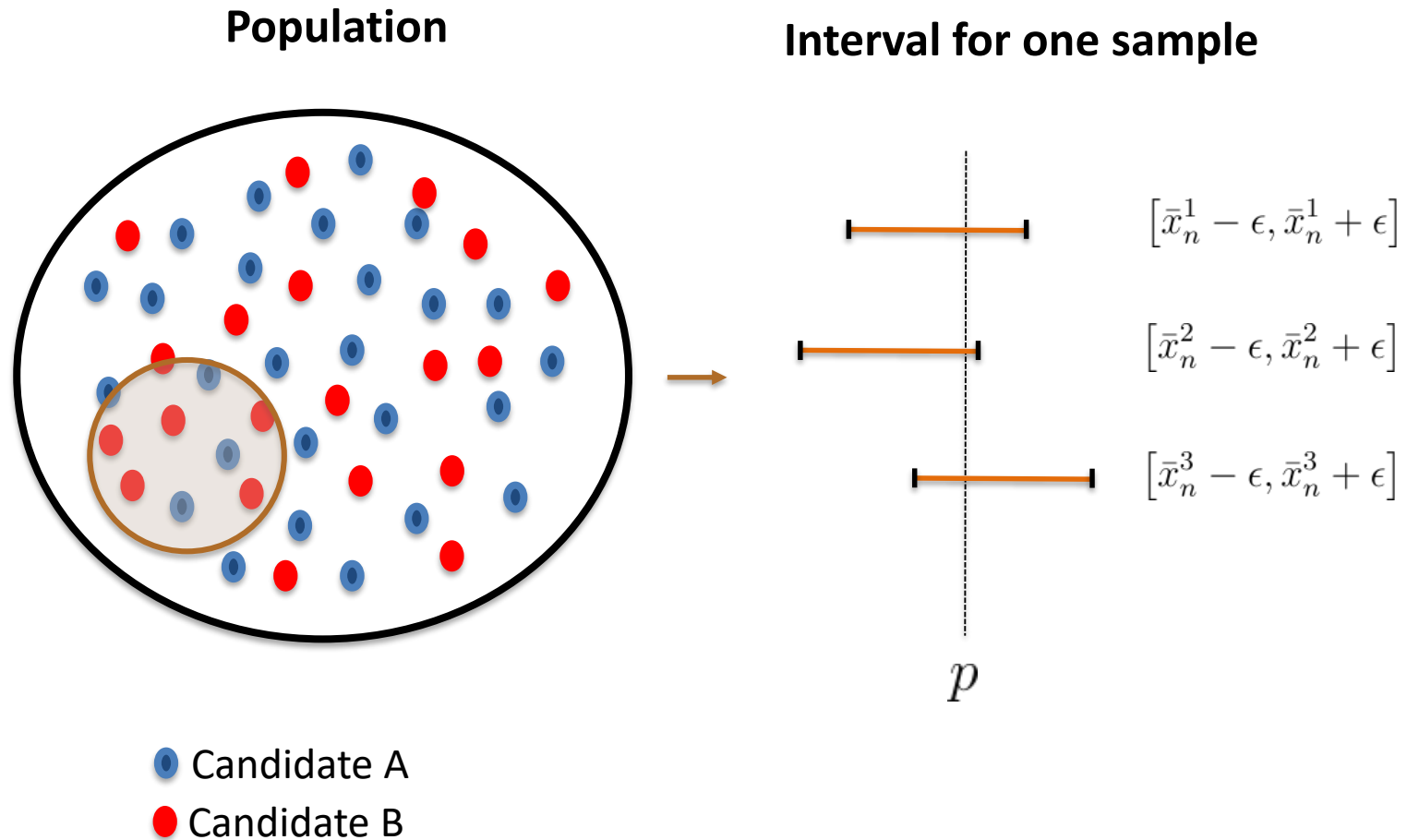
A random interval containing p



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A random interval containing p



A real situation

- Let the population size be 1.000.000
- Let the real p (that we want to estimate) be 0.5
- We choose $\epsilon = 0.03$ (3%)

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Possible situation

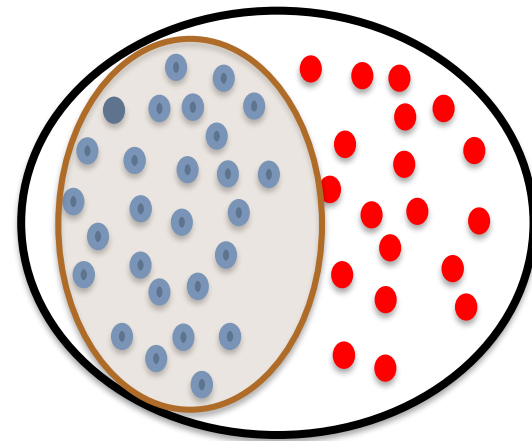
- We interview $n = 500.000$ people (that's a lot!)
- All respondents vote for Candidate A
- Then out outcome is:

$$(x_1, \dots, x_n) = (1, \dots, 1) \rightarrow \bar{x}_n = 1$$

- Then:

$$p \notin [1 - 0.03, 1 + 0.03]$$

Conclusion: Even with a 50% sample,
the margin of error is still large!



● Candidate A
● Candidate B

A probabilistic definition

- Let the population size be 1.000.000
- Let the real p (that we want to estimate) be 0.5
- We choose $\epsilon = 0.03$ (3%)

Change of perspective

Let the margin of error be defined such that, for the chosen $\epsilon > 0$:

$$P(p \in [\bar{X}_n - \epsilon, \bar{X}_n + \epsilon]) = 0.95 \quad (*)$$

Random confidence interval

We call 0.95 a *level of confidence*.

(*) reads:

If we repeat the survey 100 times, about 95 of the resulting confidence intervals contain the **real p** .

A probabilistic definition

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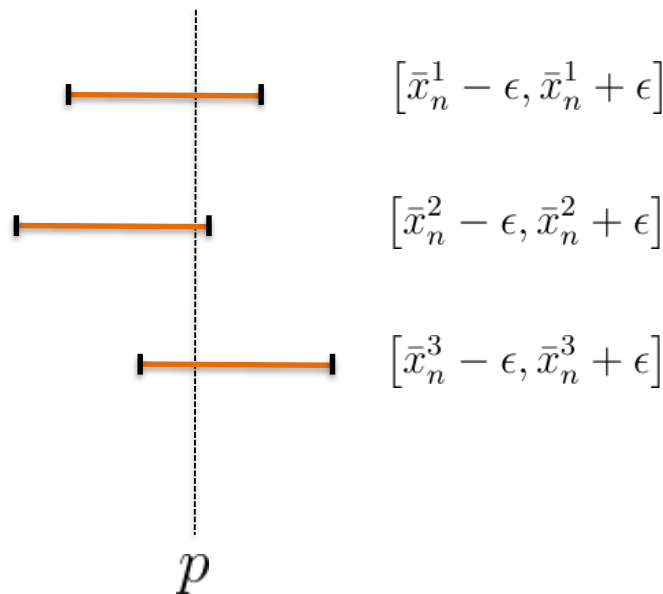
Spoiler of next slides

We will see that, for a confidence level of 95%, we need to interview fewer than 1.000 people in a population of 1.000.000 to have a margin of error $\epsilon = 0.03$ (3%).

Motivation: How reliable?

Goal: Given a **sample size** n and a level of confidence $0 \leq \alpha \leq 1$, find the margin of error $\epsilon > 0$, such that:

$$P(p \in [\bar{X}_n - \epsilon, \bar{X}_n + \epsilon]) = \text{given confidence level} = 1 - \alpha$$

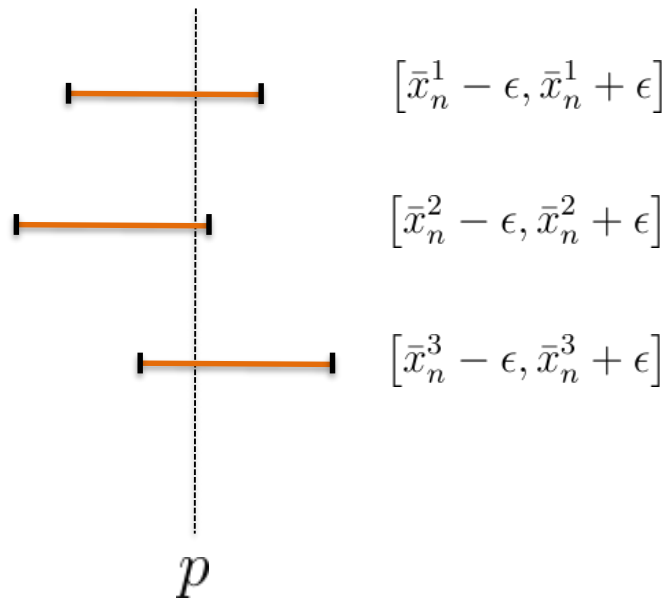


With probability $1 - \alpha$ the interval around the mean estimator that contains the real p

Motivation: How much data?

Goal: Given a **confidence level** $1 - \alpha$, for $0 \leq \alpha \leq 1$ and a margin of error $\epsilon > 0$. Find a sample size n such that:

$$P(p \in [\bar{X}_n - \epsilon, \bar{X}_n + \epsilon]) = \text{given confidence level} = 1 - \alpha$$



With probability $1 - \alpha$ the interval around the mean estimator that contains the real p

Using the symmetry of the normal

Consider $Z \sim N(0, 1)$ a normal random variable. Consider the notation:

$$q_{(1-\frac{\alpha}{2})} := Q\left(1 - \frac{\alpha}{2}\right), \quad P\left(Z \leq q_{(1-\frac{\alpha}{2})}\right) = 1 - \frac{\alpha}{2}$$

Let's compute:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right)$$

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Let's compute:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right)$$

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right) = \underbrace{P\left(Z \leq q_{(1-\frac{\alpha}{2})}\right)}_{\text{Simple equality of events}} - \underbrace{P\left(Z \leq -q_{(1-\frac{\alpha}{2})}\right)}_{\text{Definition of } q_{(1-\frac{\alpha}{2})}} = 1 - \frac{\alpha}{2} - P\left(Z \leq -q_{(1-\frac{\alpha}{2})}\right)$$

Simple equality of events

Definition of $q_{(1-\frac{\alpha}{2})}$

$$= \underbrace{1 - \frac{\alpha}{2}}_{\text{Symmetry of normal}} - \underbrace{P\left(Z \geq q_{(1-\frac{\alpha}{2})}\right)}_{\text{Sum of complementary events equals 1}} = 1 - \frac{\alpha}{2} - \left(1 - P\left(Z \leq q_{(1-\frac{\alpha}{2})}\right)\right)$$

Symmetry of normal

Sum of complementary events equals 1

$$= 1 - \frac{\alpha}{2} - \left(1 - \left(1 - \frac{\alpha}{2}\right)\right) = 1 - \alpha$$

Confidence interval

Then, we conclude that for $Z \sim N(0, 1)$:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right) = 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Consider X_i identically distributed such that $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2$. Because of the Central limit theorem, for **n large**:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \leq q_{(1-\frac{\alpha}{2})}\right) \approx 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Rearranging the inequality to isolate μ :

$$P\left(\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}\right) \approx 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Or, in the form of interval:

$$P\left(\mu \in \underbrace{\left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}\right]}_{\text{Random confidence interval}}\right) \approx 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Which reads:

Considering all possible outcomes of the random variable $\bar{X}_n$, the probability that the confidence interval contains μ is $1 - \alpha$.

Notation for confidence intervals

Confidence interval

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad \lim_{n \rightarrow \infty} P \left(\mu \in \underbrace{\left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]}_{\text{Random confidence interval}} \right) = 1 - \alpha$$

Or for “large” n :

$$P \left(\mu \in \left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right] \right) \approx 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1 \quad (*)$$

(*) reads:

Considering all possible outcomes of the random variable $\bar{X}_n$, the probability that the confidence interval contains μ is $1 - \alpha$.

Notation

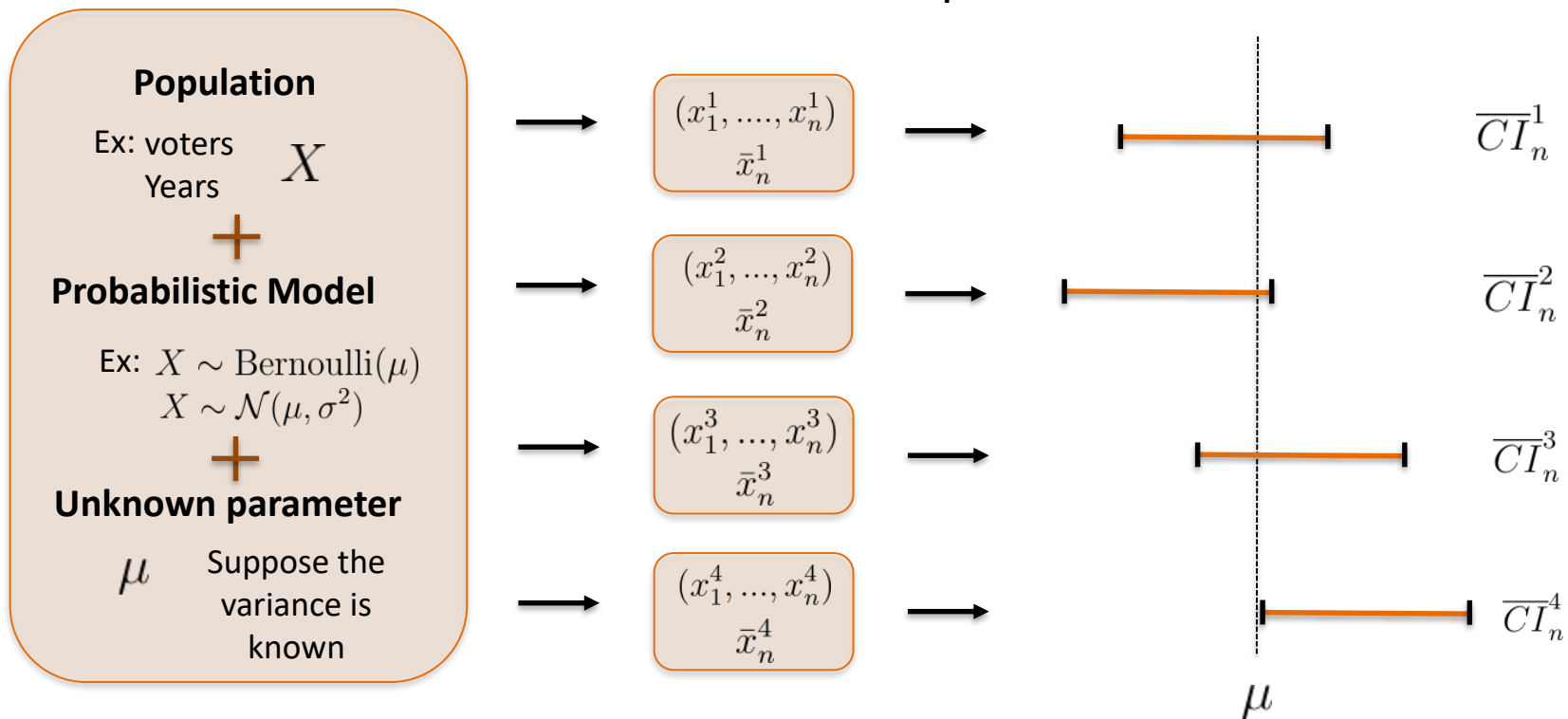
Let $(x_1, \dots, x_n)$ be an outcome of $(X_1, \dots, X_n)$, denote:

$$CI_n = \left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right] \quad \text{and} \quad \overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$$

Inference pipeline

Notation: $\overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$

Possible outcomes with n samples



Example: Calibrated measuring device

We **measure** a fixed physical quantity (for example, **temperature**) using a calibrated instrument.

The measurement error is well understood from prior calibration experiments carried out by the manufacturer. So the noise **variance** $\sigma^2 = 0.16^\circ\text{C}$ is known.

$$X_i = \text{i-th temperature measurement}$$
$$X_i = \mu + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i \in \{1, \dots, n\}$$

However, the **true value μ of the temperature being measured is unknown**.

Question: Why is the normal distribution widely used to model errors?

Statistical model

Each measurement X_i , $i \in \{1, \dots, n\}$ are independent normal random variables:

$$X_i \sim \mathcal{N}(\mu, 0.16), \quad i \in \{1, \dots, n\}$$

We want to estimate the true temperature μ

Example: Calibrated measuring device

Statistical model

Each measurement X_i , $i \in \{1, \dots, n\}$ is a normal random variable:

$$X_i \sim \mathcal{N}(\mu, 0.01), \quad i \in \{1, \dots, n\}$$

We want to estimate the true temperature μ

Data

We take $n = 25$ independent temperature measurements and obtain:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = 25^\circ\text{C}$$

Question: What is a 95% confidence interval for the true temperature μ ?

Use that, for the standard normal: $q_{0.975} = 1.96$.

Solution: Calibrated measuring device

$$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2), \quad \sigma = 0.4, \quad n = 25$$

We want a **95% confidence** interval, then: $1 - \alpha = 0.95$, and $\alpha = 0.05$

We have seen that for a standard normal:

$$P\left(-q_{(1-\frac{\alpha}{2})} \leq Z \leq q_{(1-\frac{\alpha}{2})}\right) = 1 - \alpha, \text{ for any } 0 \leq \alpha \leq 1$$

Admitting the approximation (**is it an approximation in this case?**) of the CLT, we have:

$$P\left(-q_{0.975} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq q_{0.975}\right) = 0.95, \quad q_{0.975} = 1.96.$$

We have seen that this results in a confidence interval:

$$\overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$$

Using our data:

$$\overline{CI}_{25} = \left[25 - \frac{1.96 \times 0.4}{5}, 25 + \frac{1.96 \times 0.4}{5} \right] = [24.84, 25.15]$$

With a probability of 0.95 the true temperature is in the interval $\overline{CI}_{25}$.

Simulation: Calibrated measuring device

```
mu      = 25
sigma   = 0.4
n       = 25
z       = 1.96

#number of simulations
m       = 100

halfWidth = z * sigma / sqrt(n)

count = 0

for b = 1 to m:
    generate X_1, ..., X_n as Normal(mu, sigma^2)
    Xbar = (1/n) * sum_{i=1}^n X_i

    if Xbar - halfwidth <= mu <= Xbar + halfwidth:
        count = count + 1

proportion = count / m
```

Simulation: Calibrated measuring device

Python code:

```
import numpy as np
np.random.seed(8)

# Parameters
mu = 25.0
sigma = 0.4
n = 25
alpha = 0.05
z = 1.96 # z_{0.025}

# Confidence interval that we have computed
half_width = z * sigma / np.sqrt(n)
error = half_width

#number of simulations of barX
m = 100
# Simulation
inside = 0 #start with 0 times barX inside the interval
for i in range(m):
    X = np.random.normal(mu, sigma, size=n)
    barX = np.mean(X)

    if (mu >= barX - error) & (mu <= barX + error):
        inside = inside + 1

prop_inside = inside/m

print("Half-width:", half_width)
print("Count inside:", inside)
print("Proportion inside:", prop_inside)
```

Result:

```
Half-width: 0.1568
Count inside: 96
Proportion inside: 0.96
```


Once again, back to the election poll

Election pool

A change in perspective: So far, we have visualized and analyzed some useful measures to understand data. But what if these data represent not an entire population, but only a sample?

Example: Election pool



Candidate	Number of respondents
Candidate A	362
Candidate B	638

Table 2: Election poll results with 1,000 respondents.

Statistical question: What is the probability that candidate B will win the election?

Model for election pool

Probabilistic model

Let p be the **true** (unknown) proportion of people who vote for candidate A, and $1-p$ the proportion who vote for candidate B. Based on the survey, we aim to **estimate** p .

Therefore, we have a Bernoulli random variable with parameter p :

$$X = \begin{cases} 1, & \text{if the person votes for candidate A,} \\ 0, & \text{if the person votes for candidate B.} \end{cases}$$

Data

The survey is viewed as a **realization** or **outcome** $(x_1, x_2, \dots, x_{1000})$ of the random vector $(X_1, X_2, \dots, X_{1000})$, where $X_i \sim \text{Bernoulli}(p), i \in \{1, \dots, 1000\}$

What are we assuming with this model?

- All respondents are independent
- The value of p is fixed during the survey
- Each respondent is chosen at random



Estimating p

Given a result to the election pool: $X_i \sim \text{Bernoulli}(p)$, i.i.d

$(x_1, x_2, \dots, x_n)$ which is an outcome of $(X_1, X_2, \dots, X_n)$
data set

Goal: Find the confidence interval for $n = 1000$ and a confidence level of 90%.

Information: $q_{0.95} = 1.66$

We have seen that:

$$\overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$$

Problem: The value of σ is not known and depends on p !

Confidence interval for p

Given a result to the election pool: $X_i \sim \text{Bernoulli}(p)$, i.i.d

$(x_1, x_2, \dots, x_n)$ which is an outcome of $(X_1, X_2, \dots, X_n)$
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Problem: The value of σ is not known and depends on p !

Solution: Replace σ^2 by a variance estimator:

$$\underbrace{\sigma^2 \approx \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2}_{\text{Variance estimator}} \longrightarrow \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x}_n)^2$$

Outcome from the survey

Central limit theorem + Slutsky

The theorem:

Let $X_1, X_2, \dots$ be independent and identically distributed random variables with mean $\mu := \mathbb{E}[X_1]$ and variance estimator:

$$\hat{\sigma}_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \longrightarrow \text{empirical variance}$$

Define the standardized sum:

$$Z_n^* = \frac{\sum_{i=1}^n X_i - n\mu}{\hat{\sigma}_n \sqrt{n}}$$

Then, as $n \rightarrow \infty$, the distribution of Z_n converges to the standard normal distribution:

$$Z_n^* \xrightarrow{d} \mathcal{N}(0, 1)$$

Consequence:

Let:

$$CI_n^* = \left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\hat{\sigma}_n}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\hat{\sigma}_n}{\sqrt{n}} \right]$$

Then:

$$\lim_{n \rightarrow \infty} P(CI_n^*) = 1 - \alpha, \text{ and for large values of } n: P(CI_n^*) \approx 1 - \alpha$$

Last time, back to the election poll

Confidence interval for p

Given a result to the election pool: $X_i \sim \text{Bernoulli}(p)$, i.i.d

$(x_1, x_2, \dots, x_n)$ which is an outcome of $(X_1, X_2, \dots, X_n)$
data set

Goal: Find the confidence interval for $n = 1000$ and a confidence level of 90%.
($q_{0.95} = 1.66$).

We have seen that:

$$\overline{CI}_n = \left[\bar{x}_n - q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}}, \bar{x}_n + q_{(1-\frac{\alpha}{2})} \frac{\sigma}{\sqrt{n}} \right]$$

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Solution: Replace σ^2 by a variance estimator:

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Outcome from the survey

Confidence interval for p

Goal: Find the confidence interval for $n = 1000$ and a confidence level of 90%.

$$q_{0.95} = 1.66$$

$$CI_n^* = \left[\bar{X}_n - q_{(1-\frac{\alpha}{2})} \frac{\hat{\sigma}_n}{\sqrt{n}}, \bar{X}_n + q_{(1-\frac{\alpha}{2})} \frac{\hat{\sigma}_n}{\sqrt{n}} \right]$$

- Confidence level: $1 - \alpha = 0.9, \quad \alpha = 0.1$
- Quantile: $1 - \frac{\alpha}{2} = 0.95, \quad q_{0.95} = 1.66$
- Mean estimator: $\bar{X}_n = \frac{362}{1000} = 0.362$
- Variance estimator: $\sigma^2 \approx \frac{1}{999} ((1 - 0.362)^2 \times 362 + (-0.362)^2 \times 638) = 0.231$
 $\Rightarrow \sigma \approx 0.48$
- Confidence interval: $\left[0.362 - 1.66 \frac{0.48}{\sqrt{1000}}, 0.362 + 1.66 \frac{0.48}{\sqrt{1000}} \right] = [0.33, 0.38]$

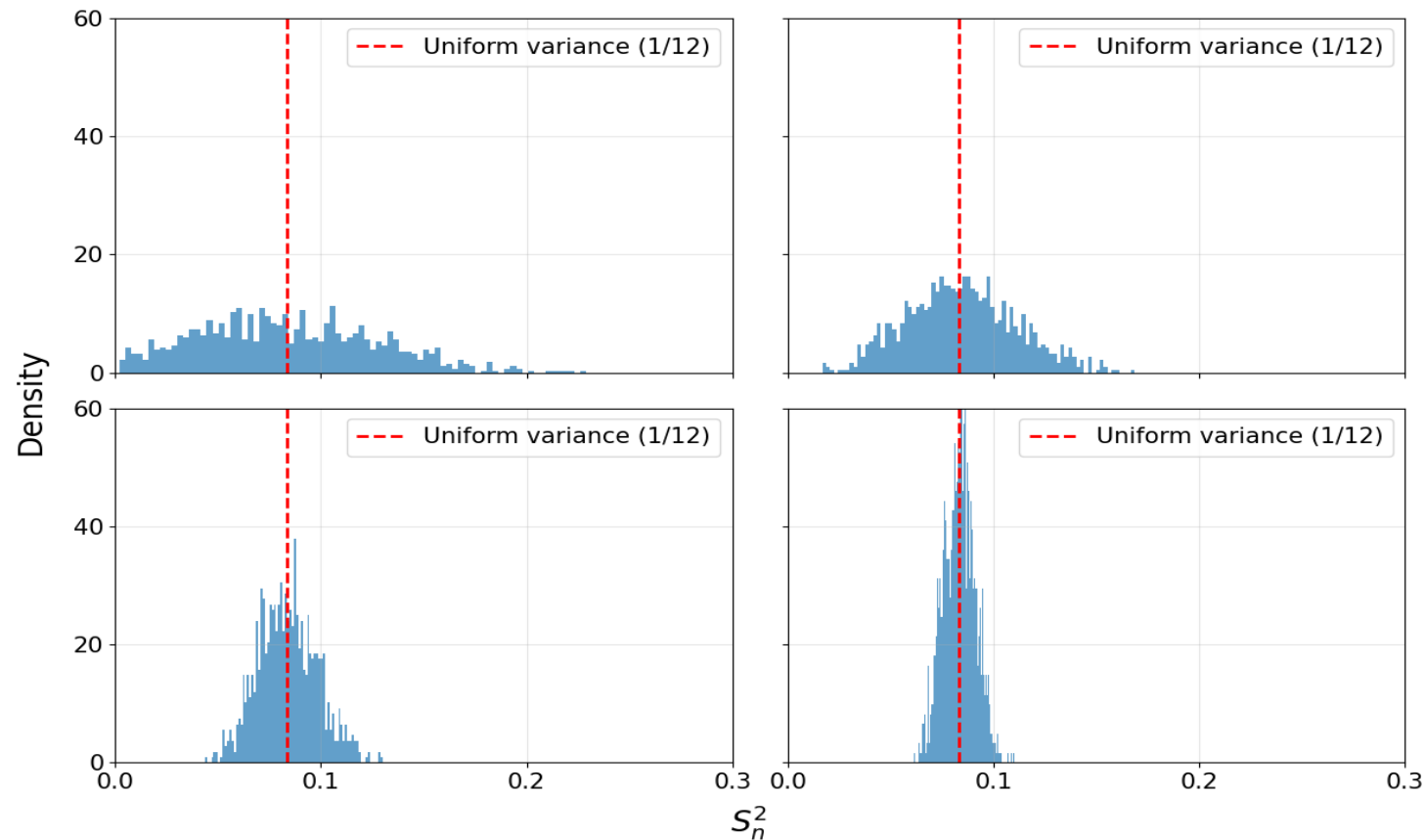
Confidence interval

Beyond the mean

Variance estimator

Example: $S_{100}^2 = \frac{1}{99} \sum_{i=1}^{100} (X_i - \bar{X})^2$, $\bar{X} = \frac{1}{n} \sum_{i=1}^{100} X_i$, for $X \sim \mathcal{U}(0, 1)$.

Observe the empirical distribution with a sample size of 1000:



Variance estimator

Example: $S_{100}^2 = \frac{1}{99} \sum_{i=1}^{100} (X_i - \bar{X})^2$, where: $\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$, for $X \sim \mathcal{U}(0, 1)$.

Discussion: Why the standard definition of this estimator has denominator $n-1$ instead of n ?

Variance estimator

Example: $S_{100}^2 = \frac{1}{99} \sum_{i=1}^{100} (X_i - \bar{X})^2$, where: $\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$, for $X \sim \mathcal{U}(0, 1)$.

Discussion: Why the standard definition of this estimator has denominator $n-1$ instead of n ?

$$\begin{aligned} \sum_{i=1}^n (X_i - \bar{X})^2 &= \sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \\ \mathbb{E} \left[\sum_{i=1}^n (X_i - \bar{X})^2 \right] &= \sum_{i=1}^n \mathbb{E}[(X_i - \mu)^2] - n \underbrace{\mathbb{E}[(\bar{X} - \mu)^2]}_{\text{Variance of the mean}} \\ &= n\sigma^2 - n \cdot \frac{\sigma^2}{n} \\ &= (n-1)\sigma^2 \end{aligned}$$

$$\text{PRECISE NUMBER} + \text{PRECISE NUMBER} = \text{SLIGHTLY LESS PRECISE NUMBER}$$

$$\text{PRECISE NUMBER} \times \text{PRECISE NUMBER} = \text{SLIGHTLY LESS PRECISE NUMBER}$$

$$\text{PRECISE NUMBER} + \text{GARBAGE} = \text{GARBAGE}$$

$$\text{PRECISE NUMBER} \times \text{GARBAGE} = \text{GARBAGE}$$

$$\sqrt{\text{GARBAGE}} = \text{LESS BAD GARBAGE}$$

$$(\text{GARBAGE})^2 = \text{WORSE GARBAGE}$$

$$\frac{1}{N} \sum (\text{N PIECES OF STATISTICALLY INDEPENDENT GARBAGE}) = \text{BETTER GARBAGE}$$

$$\left(\frac{\text{PRECISE NUMBER}}{\text{PRECISE NUMBER}} \right)^{\text{GARBAGE}} = \text{MUCH WORSE GARBAGE}$$

$$\text{GARBAGE} - \text{GARBAGE} = \text{MUCH WORSE GARBAGE}$$

$$\frac{\text{PRECISE NUMBER}}{\text{GARBAGE} - \text{GARBAGE}} = \text{MUCH WORSE GARBAGE, POSSIBLE DIVISION BY ZERO}$$

$$\text{GARBAGE} \times 0 = \text{PRECISE NUMBER}$$

Thank you for your attention and
participation!